Self-Paced Probabilistic Collaborative Representation for Anomaly Detection of Hyperspectral Images

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***Abstract*— In recent years, hyperspectral anomaly detection methods based on representation models have attracted much attention. However, when the dictionary is polluted by anomalous pixels, their performance is greatly affected. To adjust the contributions of different dictionary atoms, traditional methods usually predefine a distance weighting matrix and impose it on the dictionary matrix or coefficient vector, which may not be accurate enough. To solve this problem, a self-paced probabilistic collaborative representation detector (SP-ProCRD) is proposed in this article. It assigns weights for each atom loss term according to the probability that the pixel under test (PUT) belongs to the same class as each dictionary atom. Unlike the predefined weight matrix approach, a self-paced learning (SPL) strategy is used for iterative optimization, so that dictionary atoms participate in the representation from "good" to "bad" ones when solving the model. The representation residuals are utilized to accelerate the convergence. The proposed model can optimally represent each PUT using similar dictionary atoms and minimize the negative impact caused by anomalous atoms contained in the dictionary. In terms of weighting for SPL, an adaptive weighting scheme based on the polynomial self-paced (SP) regularizer is proposed to address the generalization issues of most previous weighting schemes. This scheme improves the generalization and automation of the model. Experimental results reveal that the proposed method produces more accurate results than existing methods and runs efficiently.**

***Index Terms*— Anomaly detection, collaborative representation (CR), hyperspectral image (HSI), self-paced learning (SPL).**

I. Introduction

HYPERSPECTRAL anomaly detection has become a booming research topic and has been successfully applied to military reconnaissance, agricultural monitoring, civil search and rescue, and many other application areas [1], [2], [3], [4], [5]. To avoid inappropriate statistical assumptions, the methods based on representation models that do not require statistical assumptions have received much attention in recent years. Of these, collaborative representation (CR) is rapidly gaining attention because of its simplicity and efficiency [6], [7], [8], [9]. Cai et al. [10] provided an innovative explanation of the effectiveness of CR in the field of pattern classification from a probabilistic point of view and proposed a probabilistic CR (ProCR) for pattern classification. The proposed ProCR has a clear probabilistic interpretation and excellent performance and has attracted much attention in the field of pattern classification [11], [12], [13], [14], [15].

Li and Du [16] first applied CR to hyperspectral anomaly detection and proposed a CR detector (CRD). However, the performance of CRD is susceptible to the interference of potential anomalous pixels in the dictionary due to the collaborative involvement of dictionary atoms [17]. In order to exclude the interference of anomalous pixels, there are some methods dedicated to constructing purer dictionaries. Su et al. [18] used principal component analysis (PCA) to extract the main information of an image and remove outliers for dictionary construction in CRD. By combining the k-nearest neighbor (KNN) method with an existing global dictionary, Wu et al. [19] proposed a nonglobal dictionary that is more reliable than the dual window dictionary. Nevertheless, constructing a completely pure dictionary without the need for prior knowledge is still very difficult. Therefore, how to minimize the impact of anomalous pixels in the dictionary on the performance of CRD is critical.

In addition to statistics and representation-based methods, there are many other methods [8]. Subspace-based methods usually project the data into a high-dimensional space [20], which can cause computational complexity. Spatial–spectral-based methods usually use a series of image processing operations such as filtering [21]. These methods require time-consuming parameterization processes. Deep-learning-based methods perform well by utilizing deep image features [22]. However, these models require a large amount of training datasets to prevent overfitting problems.

In the fields of machine learning and pattern recognition, in order to eliminate the interference of outliers, some

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researchers have introduced self-paced learning (SPL) [23]. Peng et al. [24] proposed a self-paced (SP) joint sparse representation model that can adaptively select good pixels and effectively handle the noisy neighboring pixels or spectral bands. As a weighting strategy for samples, the core design of SPL is the SP regularizer, which directly determines the optimal weights at each training period. In the original version of SPL, the training samples are usually hard/binary weighted by assigning weights "1" to the "easy" examples and "0" to the "hard" examples [25]. However, this style of hard weighting tends to lose flexibility, as it is unlikely that any two examples will be exactly equally important and learnable.

The above analysis shows that the methods based on representation models are more advantageous. Among them, CRD has received much attention due to its simplicity and efficiency. However, the detection performance of CRD is susceptible to the interference of anomalous pixels in the dictionary due to the collaborative participation of atoms in the representation of the pixel under tests (PUTs) [24]. SPL has been demonstrated to be an effective method for resisting outlier interference, but suffers from certain generalizability problems. In order to reduce the effect of anomalous pixels that may be contained in dictionary atoms and to achieve a dense representation of similar dictionary atoms, inspired by SPL, we propose an SP ProCR detector (SP-ProCRD). It uses an SP regularizer based on probabilistic subspace loss to optimize the ProCR model, while enabling dictionary atoms to progressively participate in the representation from "good" to "bad." Ultimately, it results in optimal weights and representation coefficients. To obtain a more generalized SP weighting strategy, by exploiting the robust dictionary construction method [26] and the relationship between the dictionary and the model, we propose a weighting scheme that adaptively changes the SP regularizer.

The contributions of this article are summarized as follows.

1) From the perspective of the probabilistic collaboration subspace, the probability that a PUT belongs to the same class as each atom is used to design the loss term. The representation residuals at each iteration are used to accelerate convergence.
2) Unlike the predefined weight vectors approach, SPL is used for iterative optimization, which optimizes the ProCR model while assigning weights to dictionary atoms step by step from "good" to "bad." This computational method attenuates the interference of anomalous pixels on the model, and is more accurate and stable even when the dictionary is not pure enough.
3) Aiming at poor pervasiveness of various types of previous SP regularizers, an adaptive SP weighting scheme is designed to improve the generalization and automation of the model. The scheme can adaptively adjust the discretization of the weights in SPL based on the constructed dictionary without human intervention.

The rest of the article is organized as follows. Related work is presented in Section II. Section III presents the SP-ProCRD anomaly detection method and adaptive weighting scheme. Section IV describes the experimental dataset, evaluation metrics, comparison algorithms, parameter settings, and experimental results. Section V performs a validity analysis and then discusses the effectiveness analysis and parameter sensitivity analysis of the proposed algorithm. Finally, Section VI summarizes the article.

II. Related Work

A. Probabilistic Collaborative Representation

Cai et al. [10] proposed a ProCR framework in which the probability that a testing sample belongs to the collaborative subspace of all classes can be well-defined and computed. In ProCR theory, $\boldsymbol{S}$ denotes the linear subspace spanned collaboratively by all samples in the training set $\boldsymbol{D} = [\boldsymbol{D}_1, \ldots, \boldsymbol{D}_c]$, regarded as the data matrix of the c training samples. l_X denotes the set of all labels in $\boldsymbol{X}$. Later, they proposed a ProCR-based classifier (ProCRC) that jointly maximizes the probability that each testing sample belongs to each of the classes. If $l(y) = i (i = 1, \ldots, c)$ denotes the probability of $\boldsymbol{y}$ belonging to the ith class, in order to find a common data point $\boldsymbol{Du}$ in $\boldsymbol{S}$ which maximizes the joint probability $P(l(y) = 1, \ldots, l(y) = c)$, the objective function is obtained as follows:

$$\arg\min_{\boldsymbol{u}} \|\boldsymbol{y} - \boldsymbol{Du}\|_2^2 + \alpha \sum_{i=1}^{c} \|\boldsymbol{Du} - \boldsymbol{D}_i \boldsymbol{u}_i\|_2^2 + \beta \|\boldsymbol{u}\|_2^2. \quad (1)$$

The first term $\|\boldsymbol{y} - \boldsymbol{Du}\|_2^2$ encourages the common point $\boldsymbol{Du}$ to be close to $\boldsymbol{y}$, $\|\boldsymbol{Du} - \boldsymbol{D}_i \boldsymbol{u}_i\|_2^2$ attempts to find a point close to the common point $\boldsymbol{Du}$ in the subspace of each training sample $\boldsymbol{D}_i$, and $\|\boldsymbol{u}\|_2^2$ measures the level of confidence that $\boldsymbol{Du}$ is labeled as l_X. The parameters α and β balance these three terms. The representation coefficient $\boldsymbol{u}$ can be computed as

$$\boldsymbol{u} = \left(\boldsymbol{D}^T \boldsymbol{D} + \alpha \sum_{i=1}^{c} \boldsymbol{D}_{-i}^T \boldsymbol{D}_{-i} + \beta \boldsymbol{I}\right)^{-1} \boldsymbol{D}^T. \quad (2)$$

B. Self-Paced Learning

Given the training dataset $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$, where $\boldsymbol{x}_i$ and y_i denote the ith observation sample and its label, respectively. $L(y_i, g(\boldsymbol{x}_i, \omega))$ denotes the loss function that computes the cost between the true label y_i and the predicted label $g(\boldsymbol{x}_i, \omega)$, and ω denotes the model parameter in the decision function $g(\boldsymbol{x}_i, \omega)$. The SPL [25] model consists of a weighted loss term for all the samples and an SP regularizer imposed on the weights of the samples, denoted as

$$\min_{\omega, v} \boldsymbol{E}(\omega, \boldsymbol{v}, \lambda) = \sum_{i=1}^{N} \left(v_i L\left(\boldsymbol{y}_i, g(\boldsymbol{x}_i, \omega)\right)\right) + f(\boldsymbol{v}, \lambda) \quad (3)$$

where λ is the age parameter used to control the learning pace and $f(\boldsymbol{v}, \lambda)$ denotes the SP regularizer. As the age parameter gradually increases, more samples can be automatically incorporated into the training from simple to complex in an SP manner by jointly learning the model parameter ω and potential weights $v = [v_1, v_2, \ldots, v_N]^{\mathrm{T}}$ through an iterative optimization strategy. The whole process is shown in Algorithm 1.

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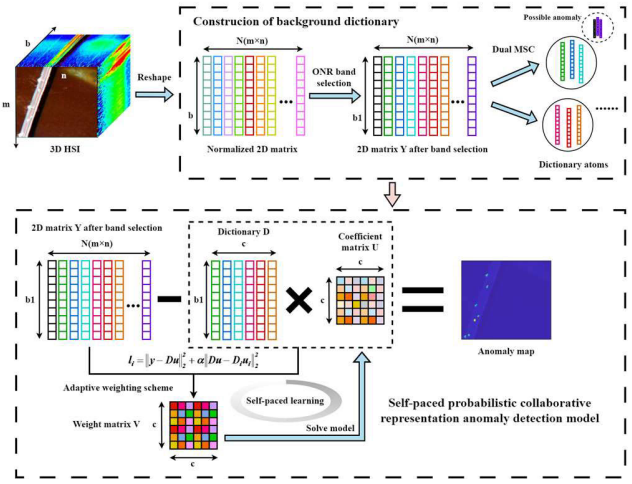


Fig. 1. Flowchart of proposed anomaly detection method. First, the original hyperspectral 3-D cube is expanded into a 2-D matrix. ONR method is utilized to select some bands. After that, a robust dictionary construction method based on Dual MSC is utilized to obtain the global dictionary. Then, the 2-D data matrix and the dictionary are input into the SP-ProCRD model. After alternating iterations through the adaptive weighting scheme and model solving, the detection result map is calculated.

III. Proposed Method

In order to more accurately adjust the contributions of different dictionary atoms when they participate in the representation, attenuate the interference of anomalous pixels on the model performance, and improve the universality of the model, this article proposes the SP-ProCRD with an adaptive weighting scheme. The main flowchart of the method is shown in Fig. 1. The main steps are as follows.

1) Band selection via optimal neighborhood reconstruction (ONR) [27] is used to improve the computational efficiency.
2) A robust dictionary construction method [26] based on dual mean shift clustering (MSC) without any prior knowledge is used to obtain a global dictionary.
3) An SP-ProCR model is developed based on the theory of CR [16] and probabilistic collaborative subspaces [10] combined with the SPL strategy [25].
4) Utilizing a new adaptive weighting scheme, suitable weight vectors are obtained adaptively.
5) The final representation residuals are calculated as detection results after iteration.

A. Self-Paced Probabilistic Collaborative Representation Detector

The key to the performance of CRD is the possible anomalous pixels in the background dictionary. However, since it is difficult to obtain a completely pure background dictionary in practice, it is usually preferred to adjust the contribution of different atoms in participating in the representation based on some distance or similarity metric. The global dictionary $\boldsymbol{D}$ in this article comes from a robust dictionary construction method without prior knowledge proposed in [26]. Referring to the theory of probabilistic collaborative subspaces, to measure the distance of each dictionary atom from the common point $\boldsymbol{Du}$, a specific probabilistic subspace loss term can be expressed as follows:

$$l_i = \|\boldsymbol{Du} - \boldsymbol{D}_i \boldsymbol{u}_i\|_2^2. \tag{4}$$

Algorithm 1 SPL Algorithm

Input: Training samples $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$, the initial model age λ, and pace size δ.
Initialize: $\boldsymbol{v} = [v_1, v_2, \cdots, v_N]^{\mathrm{T}} = 1$, $n = 1$
Repeatedly run the following steps R times:
1. update ω: $\omega = \arg\min_{\omega} \boldsymbol{E}(\omega, \boldsymbol{v})$
2. update $\boldsymbol{v}$: $\boldsymbol{v} = \arg\min_{\boldsymbol{v}} \boldsymbol{E}(\boldsymbol{v}, \omega)$
3. update $\lambda^{(n)}$: $\lambda^{(n)} = \lambda + (n-1)\delta$.
4. $n \leftarrow n + 1$
Output: ω

In order to attenuate the interference of anomalous pixels in $\boldsymbol{D}$ to the CR model and to distinguish the contribution of different dictionary atoms, inspired by the SPL strategy, we use the above loss term to assign weights to each dictionary atom. The objective function is shown as

$$\min_{u, v_i} \sum_{i=1}^{c} \left[v_i l_i + f(v_i, \lambda) \right] \tag{5}$$

where $\boldsymbol{v} = [v_1, v_2, \ldots, v_c]^{\mathrm{T}}$ denotes the weight vector of dictionary atoms, and each element $v_i \in [0, 1]$ is the weight of the loss term l_i for each corresponding atom $\boldsymbol{D}_i$. However, the weight $\boldsymbol{v}$ is monotonically increasing with respect to the model age $\lambda^{(n)}$ [25]. Since the scale of the above loss term does not fluctuate significantly during iteration, it may make the converge slowly. For this reason, we add the reconstruction term, i.e., the representation residual, to the loss term. As more dictionary atoms are involved in the representation, the representation residuals will decrease monotonically with the model age $\lambda^{(n)}$. In this way, $\boldsymbol{v}$ increases faster compared to the case when using the loss term in (4). This facilitates faster convergence. In the same model age, it is a constant that does not affect the gap between the weights of individual atoms. The improved loss term is shown as

$$l_i = \|\boldsymbol{y} - \boldsymbol{Du}\|_2^2 + \alpha \|\boldsymbol{Du} - \boldsymbol{D}_i \boldsymbol{u}_i\|_2^2. \tag{6}$$

As done in CR and ProCR, the l_2-norm constraint on the representation coefficient vector is required. In probabilistic collaborative subspace theory, it is to measure the level of confidence that the reconstructed pixels $\boldsymbol{Du}$ are included in the subspace constituted by the dictionary. Finally, we propose an SP-ProCRD model as follows:

$$\min_{u, v_i} \sum_{i=1}^{c} v_i \left(\|\boldsymbol{y} - \boldsymbol{Du}\|_2^2 + \alpha \|\boldsymbol{Du} - \boldsymbol{D}_i \boldsymbol{u}_i\|_2^2 \right) + \beta \|\boldsymbol{u}\|_2^2 + f(\boldsymbol{v}, \lambda). \tag{7}$$

The above problem can be solved using an alternative optimization strategy. Fix the weight vector $\boldsymbol{v}$, take the derivative with respect to $\boldsymbol{u}$, and set the equation to zero. Then the

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TABLE I
COMMONLY USED SP REGULARIZERS AND THE CORRESPONDING SOLUTIONS

Regularizers	$f(\boldsymbol{v},\lambda)$	v_i
Hard [28]	$-\lambda\sum_{i=1}^{c} v_i$	$\begin{cases}1, & l_i<\lambda\\ 0, & \text{otherwise}\end{cases}$
Linear [28]	$\frac{1}{2}\lambda\sum_{i=1}^{c}\left(v_i^2-2v_i\right)$	$\begin{cases}1-l_i/\lambda, & l_i<\lambda\\ 0, & \text{otherwise}\end{cases}$
Logarithmic [28]	$\sum_{i=1}^{c}\left(\zeta v_i-\frac{\zeta^{v_i}}{\log\zeta}\right)$ $\zeta=1-\lambda,\ 0<\lambda<1$	$\begin{cases}\frac{\log(l_i+\zeta)}{\log\zeta}, & l_i<\lambda\\ 0, & \text{otherwise}\end{cases}$
Mixture [28]	$-\zeta\sum_{i=1}^{c}\log\left(v_i+\frac{\zeta}{\lambda_1}\right)$ $\zeta=\frac{\lambda_1\lambda_2}{\lambda_1-\lambda_2},\ \lambda_1>\lambda_2>0$	$\begin{cases}1, & l_i\le\lambda_2\\ 0, & l_i\ge\lambda_1\\ \zeta\left(\frac{1}{l_i}-\frac{1}{\lambda_1}\right), & \text{otherwise}\end{cases}$

projection matrix can be obtained

$$\boldsymbol{M}=\sum_{i=1}^{c} v_i\left\{\sum_{i=1}^{c}\left[v_i\left(\boldsymbol{D}^T\boldsymbol{D}+\alpha\boldsymbol{D}_{-i}^T\boldsymbol{D}_{-i}\right)\right]+\beta\boldsymbol{I}\right\}^{-1}\boldsymbol{D}^T \tag{8}$$

where $\boldsymbol{I}$ denotes an identity matrix. Using $\boldsymbol{M}$, $\boldsymbol{u}$ can be obtained

$$\boldsymbol{u}=\boldsymbol{M}\boldsymbol{y}. \tag{9}$$

Fixing $\boldsymbol{u}$, the weight of each atom v_i can be obtained by solving the SP regularizer. These two optimization steps are performed alternatingly while gradually increasing the model age $\lambda^{(n)}$ to allow atoms with larger loss terms to participate in the representation. When none of the weights v_i are zero, the iteration terminates, producing the optimal representation coefficient $\boldsymbol{v}$ and weight vector $\boldsymbol{u}$. Finally, the residuals are calculated as follows:

$$r=\|\boldsymbol{y}-\boldsymbol{D}\boldsymbol{u}\|_2^2. \tag{10}$$

B. Adaptive Weighting Scheme

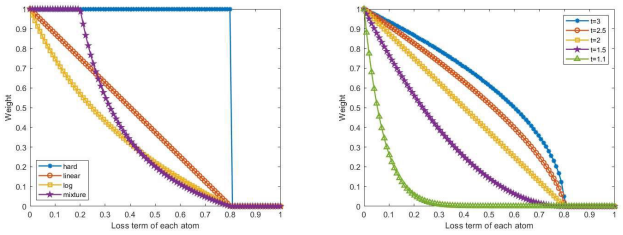
Commonly used SP regularizers are hard, linear, logarithmic, and mixed regularizers. The hard regularizer applies binary weights to the samples. However, this hard weighting tends to ignore small differences between the samples. Therefore, soft regularizers are more widely used in practice. The common SP regularizers and their corresponding closed-form solutions [28] are shown in Table I. The curves of sample weights with respect to the loss for these four different SP regularizers are shown in Fig. 2(a). Gong et al. [29] proposed a polynomial SP regularizer that approximates linear, logarithmic, and mixture regularizers well. It is defined as

$$f(\boldsymbol{v},\lambda)=\lambda\left(\frac{1}{t}\sum_{i=1}^{c}\sum_{i=1}^{c} v_i\right),\quad \lambda>0,\quad t\in\mathrm{N}^+. \tag{11}$$

The corresponding weighting solution is

$$v_i=\begin{cases}\left(1-\dfrac{l_i}{\lambda}\right)^{\frac{1}{t-1}}, & l_i<\lambda\\ 0, & \text{otherwise}\end{cases} \tag{12}$$

where t is a hyperparameter that controls the shape of the weighting function. The weighting functions corresponding to different values of t are shown in Fig. 2(b). It has a more generalized form and is able to represent more variations [30]. However, t value needs to be manually adjusted by the user based on some prior knowledge or experience value. This limits its application in different scenarios. Aiming at the problem of poor generalization of previous SP regularizers, combined with representation theory, this article proposes an adaptive weighting scheme. When the number of atoms c is large, the background will be considered to be complex and the dictionary will be overcomplete causing it to contain potentially anomalous pixels. At this point, the weights should be more discrete to distinguish the importance of individual atoms. On the contrary, when there are fewer atoms, the background is considered to be simple, the dictionary is moderate, and the weights should all be larger to ensure the ability to reconstruct all image elements. Specifically, we design the following equation:



(a) (b)

Fig. 2. Visualization of the weighting function. (a) SP regularizers' weights v_i with respect to the loss l_i in Table I. (b) Polynomial weighting regularizer with $t = 1.1, 1.5, 2, 2.5, 3$. The pace size δ is set to 0.8.

$$t=\begin{cases}2-\dfrac{c-10}{8}, & c\le 17\\ 1.1, & \text{otherwise.}\end{cases} \tag{13}$$

It can be seen that if c is large, the SP weighting function tends to be down-convex, which leads to more discrete weights; conversely, the weights are concentrated at larger values (closer to 1). For simple datasets, there are typically only about three classes of ground objects. In complex datasets, there are typically no more than 20 classes. Therefore, we use $c = 10$ as a reference to distinguish between simple and complex datasets. In particular, when $c = 10$, this scheme is equivalent to the linear soft weighting scheme. In addition, t needs a lower bound to avoid negative values for the weights. The lower bound of t is set to 1.1.

This adaptive weighting scheme improves the generalizability and automation of our proposed model. The initial model age $\lambda^{(n)}$ is set to $\lambda^{(1)}$, and the iteration pace size δ is set to $\lambda^{(1)}$. The termination criterion is set such that all atom weights are not nonzero, i.e., all atoms have been involved in the representation. The pseudocode of our proposed SP-ProCRD is shown in Algorithm 2.

IV. EXPERIMENTS

A. Comparison Methods and Evaluation Indices

In order to validate the performance of the proposed method, experiments are conducted using four different hyperspectral images (HSIs). The proposed method is compared with

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Algorithm 2 SP-ProCRD

Input: Dictionary $\boldsymbol{D}$, 2D matrix after band selection $\boldsymbol{Y} \in \mathbb{R}^{b\times N}$, regularization parameters α, β, and initial model age λ.

Initialize: $\boldsymbol{v} = [v_1, v_2, \cdots, v_c]^{\mathrm{T}} = 1$, $n = 1$.

for each PUT $\boldsymbol{y}$ **do**

While $\exists v_i = 0$ **do**

Use Eq. (13) to obtain t value;

Use Eq. (8)-(9) to update coefficient vector $\boldsymbol{u}$;

Use Eq. (12) to update the weight $\boldsymbol{v}$;

update model age: $\lambda^{(n)} = n\lambda$;

$n \leftarrow n + 1$

end while

Use Eq. (10) to calculate the anomalous level of the measured pixel.

end for

Output: Anomaly map.

seven existing representative methods, i.e., global RX (GRX) [31], local RX (LRX) [32], low rank and sparse matrix decomposition-based anomaly detection method (LRaSMD) [33], CRD [16], low rank and sparse-representation based anomaly detection method (LRASR) [34], hyperspectral anomaly detection with kernel isolation forest (KIFD) [35], and multipixel anomaly detection with unknown patterns for hyperspectral imagery (2S-GLRT) [36]. Detection accuracy is assessed by the receiver operating characteristic (ROC) [37] curve and area under the curve (AUC). The box plots are used to capture the degree of separation between anomalousities and background. The squared error ratio (SER) [38] and area error ratio (AER) [39] are also used to accurately estimate the accuracy of the algorithm.

The ROC curve depicts the relationship between the detection rate P_D and the false alarm rate P_{FA}. The SER is the sum of squares of the residuals between the true and detected values of anomalous and background pixels. The true value of anomalous pixels should be 1 and the true value of background pixels should be 0. The detected values of the anomalous and background pixels are recorded as p_i and p_j, respectively. The expression for SER is given by

$$\mathrm{SER} = \frac{\sum_{i=1}^{n_{\mathrm{anomaly}}(\mathrm{GT})} (p_i - 1)^2 + \sum_{j=1}^{n_{\mathrm{bck}}(\mathrm{GT})} (p_j - 0)^2}{N_p} \times 100 \tag{14}$$

where $n_{\mathrm{anomaly}}(\mathrm{GT})$ is the number of true anomalous pixels, $n_{\mathrm{bck}}(\mathrm{GT})$ is the number of true background pixels, and N_p is the total number of pixels. The smaller the SER, the more accurate the result is.

AER evaluates the degree of separability between anomalous pixels and the background. The y-axis represents the P_D and P_{FA} obtained at different thresholds, and the x-axis represents the normalized threshold. The expression for the AER is given by

$$\mathrm{AER} = \frac{1 - \mathrm{AP}_{\mathrm{FA}}}{1 - \mathrm{AP}_D} \tag{15}$$

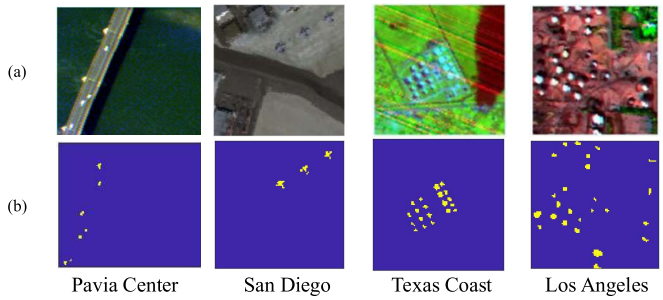


Fig. 3. (a) Pseudocolor images. (b) Reference maps of anomalous pixel distribution.

where $\mathrm{AP}_{\mathrm{FA}}$ and AP_D are the areas under the curves P_{FA} and P_D, respectively. Larger AER values imply better detection performance.

B. Experimental Datasets

The first dataset is a scene with small targets and a simple background. The image was acquired by the reflectance optical spectral imaging system (ROSIS) sensor at the Pavia Center with a spatial resolution of 1.3 m. A sub-image of size 105 × 120 with 102 bands was selected for the experiment. The background is mainly bridges, rivers, bare soil, and buildings. The anomaly targets are vehicles on bridges consisting of 43 pixels. The pseudocolor image and the anomaly reference map are shown in Fig. 3.

The second dataset is a scene with large targets and a complex background. The scene was acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) located in San Diego, CA, USA. In this article, a sub-image of size 100 × 100 with 186 bands is selected for the study. The scene consists mainly of buildings with different roofs, tarmacs with different materials, an airport runway, and a small amount of vegetation. 57 pixels are selected as anomalies, including full-pixel anomalies for the aircraft bodies and sub-pixel targets for the edges of the aircraft. The pseudocolor image and the anomaly reference map are shown in Fig. 3.

The third dataset is a dense target scene with a simple background. It was obtained by imaging the Texas coast using AVIRIS. The image contains 207 bands. It has a spatial size of 100 × 100 and a spatial resolution of 17.2 m. The background is the Texas coast image scene. The anomalous targets are buildings on the coast. Fig. 3 shows the pseudocolor map and the anomaly reference map.

The fourth dataset is a multitarget scene. It was obtained by imaging Los Angeles with AVIRIS with a spatial resolution of 7.1 m. The image used in the experiment contained 100 × 100 pixels, and 205 bands were retained after removing low-quality bands. The background is the Los Angeles image scene. The anomalous targets are buildings. The pseudocolor image and anomaly reference map are given in Fig. 3.

C. Parameter Setting

The parameters involved in the dictionary construction of the proposed anomaly detection method include the number of selected bands $b1 = 10$ and the range of the second bandwidth

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TABLE II
AUC, SER, AER, AND RUNNING TIME OF EACH ALGORITHM FOR FOUR DATASETS

Datasets		SP-ProCRD	SP-ProCRD-LS	RX	LRX	LRaSMD	CRD	LRASR	KIFD	2S-GLRT
Pavia Center	AUC	**0.9996**	0.9990	0.9982	0.9649	0.9517	0.9944	0.9991	0.8500	0.8958
	SER	**0.1435**	0.2288	0.3004	0.4293	0.2124	0.9926	0.9891	4.2553	0.4323
	AER	**1.9547**	1.3490	1.1895	1.1136	1.6016	1.6427	1.3418	1.3698	1.2600
	Time(s)	2.19	2.20	**0.04**	27.11	5.86	100.67	195.63	25.19	32.54
San Diego	AUC	**0.9919**	0.9915	0.9090	0.8734	0.9840	0.9844	0.9758	0.9807	0.7183
	SER	1.1378	1.3492	0.7238	0.5932	**0.5669**	0.7954	0.8941	1.9085	1.2514
	AER	1.4595	1.4699	1.0334	1.0892	1.2937	1.1881	0.9969	**1.6715**	1.0193
	Time(s)	0.68	0.78	**0.31**	59.86	9.08	188.95	48.01	32.55	376.17
Texas Coast	AUC	**0.9962**	0.9955	0.9946	0.8969	0.9340	0.9484	0.8071	0.8325	0.8060
	SER	1.1289	2.4441	1.2834	1.4787	**1.0412**	1.4031	1.4384	3.7449	1.6085
	AER	1.3738	**1.3943**	1.1109	1.0269	1.2697	1.0719	1.0499	1.3198	1.0218
	Time(s)	1.54	1.68	**0.18**	59.54	10.29	85.41	41.27	22.93	72.30
Los Angeles	AUC	**0.9962**	0.9958	0.9887	0.8323	0.9617	0.9719	0.9491	0.9791	0.9050
	SER	**2.0323**	2.0571	2.3508	2.6899	2.4687	2.7249	2.3006	2.5191	2.7768
	AER	1.1647	1.1596	1.0814	1.0068	1.0525	1.0264	1.0279	**2.0050**	1.0491
	Time(s)	1.16	1.66	**0.34**	54.11	9.83	126.96	40.77	15.68	68.74

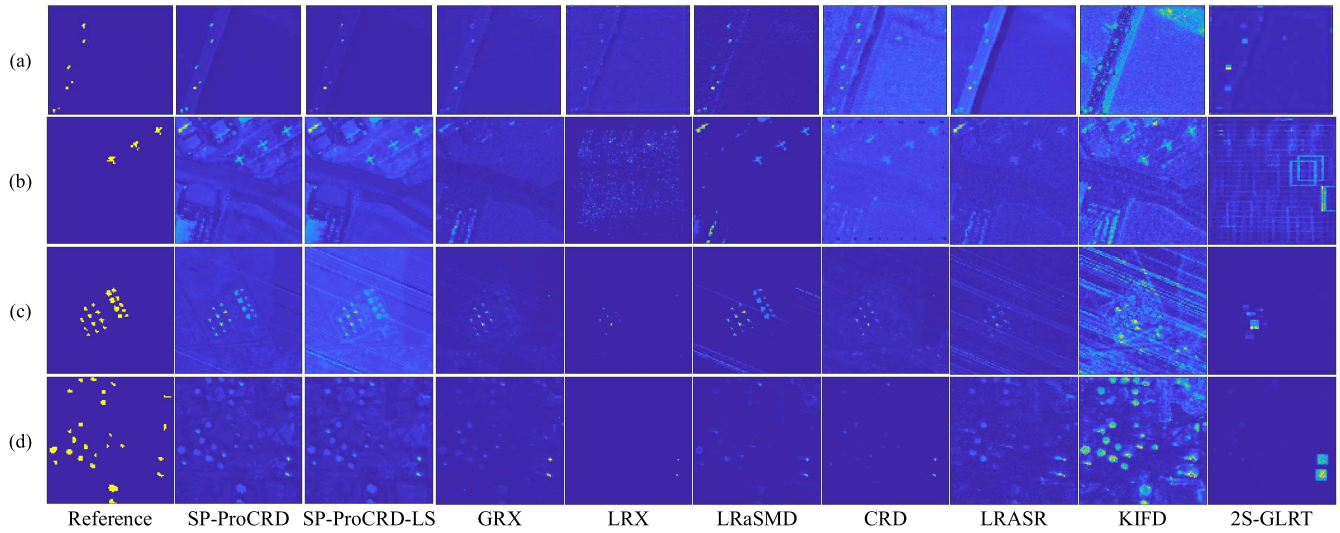


Fig. 4. Anomaly maps of SP-ProCRD, SP-ProCRD-LS, GRX [31], LRX [32], LRaSMD [33], CRD [16], LRASR [34], KIFD [35], and 2S-GLRT [36]. (a) Pavia center. (b) San Diego. (c) Texas Coast. (d) Los Angeles.

[0.1, 0.3]. The parameters involved in the SP-ProCRD model include α, β, and λ. The large range of the parameters α and β is set to [1e-6, 1e1]. The range of parameters λ is set to [1e-3, 1e1]. In the comparison algorithm, the range of dual window sizes for LRX and CRD is set from 3 to 23 for the inner window and from 5 to 25 for the outer window. The best result within the range of parameter variations is considered as the final result. In this article, the parameters of LRaSMD, CRD, LRASR, KIFD, and 2S-GLRT are set as recommended values. In addition, in order to evaluate the effectiveness of the proposed model and adaptive weighting scheme, comparative experiments are conducted using the CRD model, ProCR model, and SP-ProCRD using linear soft weighting (SP-ProCRD-LS). In this experiment, the dictionary construction conditions are the same.

D. Results

The anomaly detection maps are shown in Fig. 4. The AUC, AER, SER, and running time of each algorithm are shown in Table II. In order to visualize the experimental results, the optimal detection results are highlighted in bold and the suboptimal detection results are underlined. The ROC curve is shown in Fig. 5(a). The degree of separation between anomaly and background for the nine algorithms for the four datasets is discussed in Fig. 5(b). The Pavia Center dataset has small anomaly targets and simple backgrounds that are easy to detect. As can be seen from the figure, SP-ProCRD is the best at highlighting anomalies, has the best background anomaly separation, and is second only to LRaSMD in suppressing the background. From the ROC curve, it can be seen that SP-ProCRD is significantly closer to the top left corner. By quantitative analysis, the proposed method is optimal on AUC, SER, and AER, and the operational efficiency is second only to GRX, which indicates that SP-ProCRD performs significantly better than the existing methods on this type of dataset.

The San Diego dataset has unremarkable anomaly targets, a complex background, and high detection difficulty. In Fig. 4, the targets of GRX, LRX, and 2S-GLRT are almost invisible,

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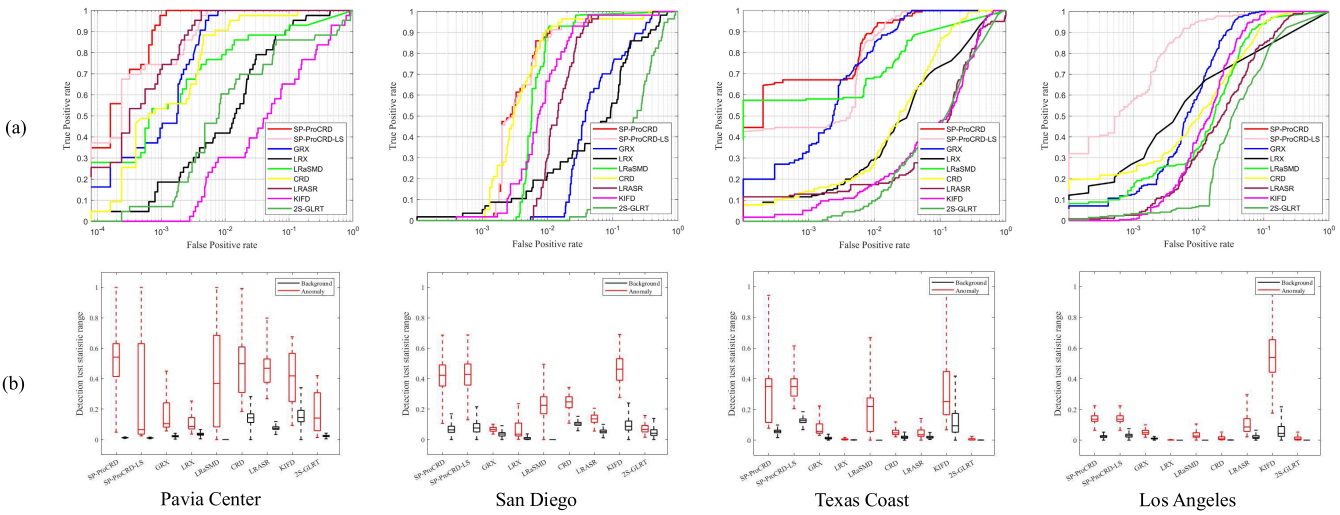


Fig. 5. ROC curve and box-plots of several anomaly detection methods for the four datasets. (a) ROC curve. (b) Box plots.

and the targets of CRD and KIFD are the most prominent, but they are not well separated from the background. Quantitatively analyzed by the four evaluation metrics, SP-ProCRD has the best AUC and SP-ProCRD-LS is the second best. For the other three metrics, SP-ProCRD-LS has the second-best AER after KIFD, and SP-ProCRD has the second-best operational efficiency after RX. In summary, the proposed method offers a slight advantage in this dataset.

The Texas Coast dataset has a simple background with large and small anomalous targets and a concentrated distribution. As can be seen from the figure, only SP-ProCRD, SP-ProCRD-LS, and LRaSMD can detect clear and complete anomaly pixels. However, SP-ProCRD-LS and LRaSMD are poor in background suppression and the detection result maps are pure enough, while SP-ProCRD can completely separate the anomaly from the background. As shown in Table II, SP-ProCRD has the highest AUC, and the other three metrics are second best, with AER slightly lower than SP-ProCRD-LS.

The Los Angeles dataset has many anomaly targets with different sizes and uneven distribution, making detection difficult. As can be seen from the figure, the ROC curves of SP-ProCRD and SP-ProCRD-LS are obviously more skewed to the upper left corner, and they look completely overlapped. This is due to the fact that the background is very homogeneous, and under the condition that the dictionaries are identical, the representation model plays a major role, with the different weighting schemes reflecting subtle disparities. As shown in Table II, SP-ProCRD has the best AUC and SER, the suboptimal AER, and operational efficiency, and all four metrics are slightly better than SP-ProCRD-LS.

TABLE III
COMPARISON OF SP-PROCRD WITH THREE MODELS ON FOUR DATASETS

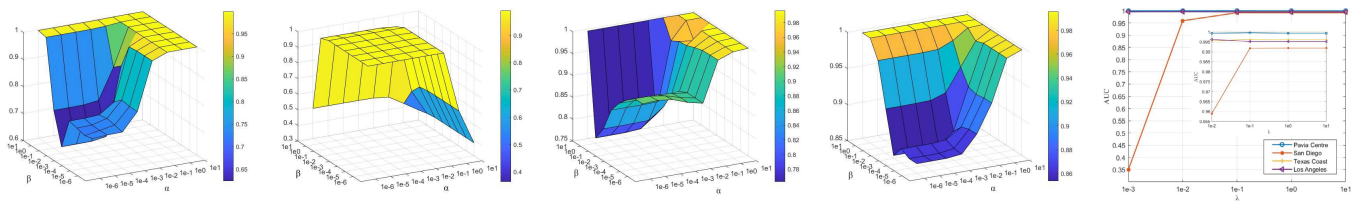
Datasets		SP-ProCRD	SP-ProCRD-LS	ProCR	CRD
Pavia Center	AUC	**0.9996**	0.9990	0.9985	0.9664
	SER	**0.1435**	0.2288	0.9604	0.4625
	AER	**1.9547**	1.3490	1.2137	1.0000
	Time(	2.19	2.20	**1.21**	1.93
San Diego	AUC	**0.9919**	0.9915	0.9919	0.9910
	SER	1.1378	1.3492	1.0588	**0.8306**
	AER	1.4595	**1.4699**	1.4242	1.2674
	Time(	0.68	0.78	**0.64**	0.97
Texas Coast	AUC	**0.9962**	0.9955	0.9958	0.9274
	SER	**1.1289**	2.4441	1.5338	1.3475
	AER	1.3738	**1.3943**	1.2141	1.0265
	Time(	1.54	1.68	**0.75**	1.21
Los Angeles	AUC	**0.9962**	0.9958	0.9957	0.8626
	SER	**2.0323**	2.0571	2.1221	2.6156
	AER	**1.1647**	1.1596	1.1473	1.0190
	Time(	1.16	1.66	**0.76**	0.94

V. DISCUSSION

A. Validity Analysis

1) Validity Study of SPL: In order to demonstrate the validity of the proposed method, CRD and ProCR are used to compare with the proposed SP-ProCRD and SP-ProCRD-LS. To make the experimental results convincing, it is ensured that the dictionary construction conditions are identical. As shown in Table III, the proposed method has higher detection accuracy and stability for the four datasets on the four evaluation metrics. For the Pavia Center and Los Angeles, the performance of each anomaly detection model is in the following order: SP-ProCRD>SP-ProCRD-LS>ProCR>CRD. However, for the San Diego dataset, the detection performance of SP-ProCRD is the same as that of ProCR, and both achieve good detection. This is because the dataset has a simple background distribution and small anomaly targets for easy detection. For the Texas Coast dataset, the performance of each anomaly detection model is in the following order: SP-ProCRD>ProCR>SP-ProCRD-LS>CRD. The above analysis shows that the proposed SP-ProCRD is applicable to both simple scene images and complex scene images, demonstrating that it is very generalizable and stable.

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(a) (b) (c) (d) (e)

Fig. 6. AUC of SP-ProCRD with the change of α, β, and λ. (a) Pavia center, (b) San Diego, (c) Texas coast, and (d) Los Angeles show the AUC of SP-ProCRD with the change of α, β. (e) AUC of SP-ProCRD with the change of λ on four datasets.

2) Effectiveness of Adaptive Weighting: In order to demonstrate the effectiveness and robustness of the proposed adaptive weighting scheme, in this article, we use SP-ProCRD to detect anomaly detection in the four datasets in comparison with an SP probabilistic CR model using linear soft weighting. The dictionary construction conditions and all parameters are exactly the same, and the best results from the experiments are chosen as the final results, as shown in Fig. 4. The results of the comparison on the four evaluation indicators are presented in Table II. For all four different datasets, the adaptive weighting proposed in this article has some advantages. The detection accuracy is improved by 0.06%, 0.04%, 0.07%, and 0.04%, respectively. This is because, in complex scenes, representative dictionary atoms are selected for representation by discriminative weighting. Moreover, in simple scenes, the adaptive weighting scheme is better at suppressing the background and highlighting anomalous pixels.

B. Parameter Analysis

The parameters involved in this article include the model parameters α, β, and λ. For the four datasets, the number of bands $b1$ selected by ONR is set to 10. In dual MSC, the first bandwidth is preset by the k-dist method. For the Pavia Center, Texas Coast, and Los Angeles datasets, the first bandwidths are set to 0.3, 0.4, and 0.5, respectively. The second bandwidths are all set to 0.1. For the San Diego dataset, the first bandwidth is set to 0.5, and the second bandwidth is set to 0.28. Fixing λ, the AUC of the SP-ProCRD for the four datasets when α and β are varied within that range is shown in Fig. 5.

For the Pavia Center dataset, significant changes in anomaly detection occur when $\alpha <$ 1e-1 and $\beta <$ 1e0. In the other ranges, the AUC was quite high and stable in change. The AUC was highest when $\alpha >$ 1e-1 and β took any value in the range. For the San Diego dataset, when $\alpha <$ 1e0 and $\beta <$ 1e0, the detection accuracy is stabilized against changes in these two parameters. On fine-tuning the parameters, it can be noticed that the AUC reaches its maximum when both α and β are taken as 1e-2. For the Texas Coast dataset, when $\alpha <$ 1e-1, the detection accuracy is sensitive to changes in both parameters. When $\alpha >$ 1e-2, the AUC fluctuated slightly around the highest value. After fine tuning it can be obtained that AUC reaches its maximum when α takes 1e-1, β takes 1e1. For the Los Angeles dataset, detection accuracy is sensitive to changes in both parameters when $\alpha <$ 1e0 and $\beta <$ 1e-1. In the other ranges, the AUC varied more steadily. The AUC was highest when α was taken as 1e1 and β was taken as 1e-1.

Fixing α and β, the AUC of SP-ProCRD for the four datasets when the range of λ is taken as [1e-3, 1e1] is shown in Fig. 6(e). It can be seen that the AUC $\geq$ 0.95 when $\lambda \geq$ 1e-2. In particular, the AUC of the proposed algorithm approximates the optimal result when $\lambda \geq$ 1e-1. Further observation leads to the conclusion that for Pavia Center and Los Angeles datasets, the AUC exhibits a trend of increasing and then decreasing as λ increases monotonically. For the other two datasets, the AUC tends to increase and then remain unchanged. After fine-tuning, the AUC reaches its maximum when lambda is taken as 1e-1, 1e0, 1e0, and 1e-2 on the four datasets in order.

To sum up, the proposed SP-ProCRD has good universality and robustness for different scenarios.

VI. Conclusion

In this article, a new SP ProCR hyperspectral anomaly detector is proposed. The CR anomaly detection method is improved in terms of both model optimization and adaptive weighting scheme. The effect of anomalous pixels on the performance of the representation model is weakened by SPL, and the reconstruction error at each iteration is utilized to accelerate the convergence. The improvement of the adaptive weighting scheme takes into account the discriminative nature of the weights and the stability of the model in different cases and improves the generalizability and automation of the model. Experiments on four real hyperspectral datasets demonstrate that the proposed method outperforms existing algorithms in terms of both accuracy and efficiency.

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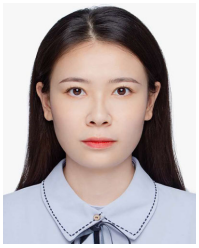
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